

Local Session Synthesizer & Knowledge Graph Builder

Target Stack: Screenpipe Engine + Tauri **Integration:** Obsidian Vault Sync Wrapper

Privacy Model: 100% Local-First

1. Executive Summary & Value Proposition

This document establishes the architecture and product positioning for a highly efficient, lightweight productivity application. The tool programmatically tracks user activity (screen OCR, user audio, and UI-accessibility structure) strictly during bounded, user-initiated deep work sessions. Upon completing a session, it constructs a dense knowledge map, synthesizes contextual connections, cross-references recent inbox communication, and generates highly organized, interconnected markdown documents natively exported into the user's Obsidian Vault.

Core Value Proposition: "Turn your focused deep-work blocks into fully contextualized, automatically cross-linked Obsidian knowledge networks. Zero cloud reliance, absolute data autonomy, and zero idle-resource overhead."

2. Market Landscape & Niche Analysis

Market Background & Evolutionary Shifts

- **Rewind AI & Limitless AI:** Pioneered 24/7 background recording but faced steep consumer pushback regarding battery optimization, localized hardware drag, and persistent security anxieties. Their pivot into cloud-based audio pendants and eventual industry consolidation validated the raw value of human digital context data but highlighted a fatal friction point: tech-savvy power-users reject perpetual, external cloud tracking.
- **Native Alternatives (Microsoft Windows Recall):** Native OS integrations introduce broad, uncured tracking layers that exacerbate privacy concerns and lack open-source customization. They remain generalized timeline search tools rather than structured synthesis engines.

The Defensible Market Niche

A massive commercial window exists for high-intent information professionals (developers, academic researchers, digital archivists, and the core Obsidian community) who require automated document creation but aggressively safeguard their data. By building an application that operates strictly on-demand (Session-Based) and preserves local data sovereignty, this tool neutralizes "surveillance anxiety" and transforms a raw data dump into synthesized intellectual capital.

3. Product Core Architecture

The system splits functionality into an event-driven session framework and a decoupled asymmetric background routine to optimize compute resources.

Component	Functional Responsibility	Technical Mechanics
Session Tracker	Active tracking of screens, text OCR, and voice audio streams <i>only</i> while the session timer runs.	On-demand activation of underlying screen/audio pipes via explicit programmatic triggers.
Inbox Summarizer	Passive background tracking of high-level communication streams to feed cross-context synthesis.	Ultra-lightweight background cron task pulling IMAP headers or API metadata every 30-60 mins.
Synthesis Engine	Bundles timeline-matched audio transcripts and textual context, formatting them for knowledge graphs.	Structured prompt engineering executed via local LLM or securely isolated Cloud APIs.

4. Lightweight Execution via Open Source (Screenpipe Integration)

To bypass engineering overhead associated with low-level OS drivers, kernel extensions, and cross-platform capture optimization, the application utilizes the **MIT-licensed, open-source core framework of Screenpipe** as a headless utility.

Programmatic Process Control

Unlike Screenpipe's default 24/7 logging configuration, this application operates the pipeline as an episodic worker, yielding major platform efficiencies:

- **0% Idle CPU Utilization:** Screenpipe’s capture engines are completely terminated when a user ends a session, leaving zero continuous battery drain or background processing overhead.
- **High-Signal Context Filtering:** Eliminates data noise (e.g., casual browsing, messaging apps, unrelated tabs). Every byte of captured data maps directly to active session windows, optimizing context-window usage and drastically driving down token costs or local inferencing cycles.
- **Technical Hooking:** The custom application framework wraps Screenpipe's CLI. On user toggle, it executes command-line binaries such as `screenpipe --start-audio --ocr`, streaming temporary data records to a discrete local SQLite database, and calling an abrupt termination signal on session close.

5. The Synthesis and Export Loop

The unique value add of this software relies on its custom interpretation of raw timelines into structural nodes. The export workflow operates under specific algorithmic conditions:

1. **Temporal Context Pairing:** The engine pairs audio text with concurrent screen states. For example, if a user verbalizes an intention ("I need to refactor this class structure") while viewing an exact lines-of-code sequence, the engine matches the timestamp to associate the vocal action-item directly with the technical code content.
2. **Cross-Context Convergence:** Right before finalizing the summary, the engine interrogates the Inbox cache. If an unread communication artifact links directly to the concept analyzed in the session, it injects an immediate cross-reference alert (e.g., "[!] High-Priority Connection: E-mail received from Product Lead regarding this specific module at 10:42 AM").
3. **Obsidian-Native Formatting:** The final pipeline step generates raw Markdown (.md files) writing directly into a designated local directory. The layout uses Markdown headers and automatically inserts structural semantic markers ([[Wikilinks]]) by identifying repetitive metadata keywords, mapping the session cleanly into an existing Obsidian knowledge graph.

6. Commercialization & Legal Guardrails

- **Open-Source Licensing Compliance:** Because Screenpipe core components utilize the permissive MIT License, the proprietary source code of this wrapper application, UI dashboard, and synthesis prompts can be safely distributed commercially under closed licenses. Enterprise Edition folder elements (ee/) within the open-source repo must be strictly avoided.
- **Monetization Strategies:**
 - *Tier 1: Bring-Your-Own-Key / Local Only.* Single upfront software fee. Users leverage local Ollama nodes or supply private LLM API tokens. Zero marginal maintenance costs to the business.
 - *Tier 2: Subscription Premium Suite.* Recurring SaaS model providing optimized cloud API tokens (Claude/GPT) for zero-latency, cloud-computed summarizations without local hardware strain.